fastsae: Frequentist and Bayesian Small Area Estimation in R

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Abstract Small area estimation (SAE) combines survey data with auxiliary information to estimate parameters of domains whose samples are too small for direct estimation. We present fastsae, an R package that brings the main area-level and unit-level models of official statistics together in one interface. The frequentist part fits the Fay-Herriot model, its spatial and spatio-temporal extensions, the Battese-Harter-Fuller unit-level model and the two-fold nested error model, with analytical and bootstrap mean squared errors, in compiled code that never forms a dense inverse of the marginal covariance matrix. The hierarchical Bayes part fits area-level, unit-level and two-fold models with Gaussian, beta, gamma, binomial, Poisson and negative binomial likelihoods and spatial, temporal and space-time random effects, by integrated nested Laplace approximations or, through the same call, by Hamiltonian Monte Carlo in JAX. For beta, gamma and negative binomial data the dispersion of each area is fixed by the design-based sampling variance of its direct estimate, and the default priors were chosen by simulation and are documented. Further functions benchmark estimates to published totals at one or two levels, classify them by relative standard error, compare models, and produce maps and spreadsheet exports. We describe the methods and their implementation, compare runtimes and estimates with the packages sae, emdi and tipsae, report a design-based simulation, and illustrate a complete workflow with poverty rates of the health districts of Emilia-Romagna.

1 Introduction

Official statistics are increasingly asked for indicators at levels of geography or population detail that national surveys were not designed for: poverty rates of districts, unemployment of municipalities, or health indicators of subregions (Rao and Molina, 2015; Pfeffermann, 2013). Direct estimates from such domains rest on few observations and have large sampling variances. Small area estimation (SAE) reduces these variances by linking the domains through a model with auxiliary information, at the area level when only domain aggregates are available (Fay and Herriot, 1979) and at the unit level when auxiliary variables are known for the sampled units and as population means (Battese et al., 1988). The basic models have been extended to spatially correlated area effects (Petrucci and Salvati, 2006; Pratesi and Salvati, 2008), to repeated surveys (Marhuenda et al., 2013), and, for proportions, rates and counts, to non-Gaussian likelihoods, either through empirical best predictors of generalized linear mixed models (Boubeta et al., 2016) or by hierarchical Bayes (HB) estimation, which also accounts for the uncertainty about the variance parameters (Rao and Molina, 2015; Janicki, 2020; Fabrizi et al., 2011).

Producing small area statistics routinely raises three practical requirements. First, the methods must scale to the thousands of areas, and the repeated fits of bootstrap mean squared error (MSE) estimation, of national production systems. Second, the HB models that suit bounded and count indicators should not require each analyst to write and diagnose an MCMC sampler. Third, published small area estimates are usually benchmarked to direct estimates of larger domains and accompanied by indicators of their reliability, so these steps belong in the same workflow as the model fit.

Several R packages implement parts of this (Table 1). **sae** (Molina and Marhuenda, 2015) covers the frequentist area-level models, including spatial and spatio-temporal Fay-Herriot models, and the unit-level model in R code. **emdi** (Kreutzmann et al., 2019) adds transformations, the empirical best predictor for poverty indicators, benchmarking to a national total, maps and spreadsheet export. **hbsae** (Boonstra, 2022) fits Gaussian area- and unit-level HB models by numerical integration over the variance parameter and benchmarks estimates to any set of linear constraints. **tipsae** (De Nicolò and Gardini, 2024) fits beta-type area-level models with spatial and temporal effects in **Stan**, **mcmcsae** (Boonstra, 2026) fits a wide class of multilevel models by MCMC, and **SUMMER** (Li et al., 2019, 2025) fits area- and cluster-level models with INLA, focusing on demographic and health surveys.

fastsae covers these models and steps in one interface and was written with the first two requirements in mind. Its contributions are:

1. Compiled implementations of the Fay-Herriot, spatial, spatio-temporal, unit-level and two-fold models with analytical and bootstrap MSEs that never form a dense inverse of the marginal covariance matrix, and run the bootstrap in parallel with results that do not depend on the number of threads.
2. One HB interface, `hb_area()`, `hb_unit()` and `hb_twofold()`, for six likelihoods and spatial, temporal and space-time effects, fitted by INLA (Rue et al., 2009) or by the No-U-Turn sampler

Table 1: Small area estimation packages for R. EBLUP: empirical best linear unbiased predictor; EB(P): empirical best (predictor); HB: hierarchical Bayes; HMC: Hamiltonian Monte Carlo. Taken from the package documentation (sae 1.3, emdi 2.2.3, hbsae 1.2, tipsae 1.0.3, mcmcsae 0.8.2, SUMMER 2.0.0).

Feature	fastsae	sae	emdi	hbsae	tipsae	mcmcsae	SUMMER
Inference	REML/ML, HB (INLA, HMC)	REML/ML	REML/ML	REML, HB (num. integration)	HB (Stan HMC)	HB (Gibbs, HMC)	HB (INLA)
Area-level model	yes	yes	yes	yes	beta-type	yes	yes
Spatial / spatio-temporal EBLUP	SAR / SAR + AR(1)	SAR / SAR + AR(1)	SAR / –	–	–	–	–
Unit-level model	EBLUP, HB	EBLUP, EB	EBLUP, EBP	EBLUP, HB	–	HB	HB
HB likelihoods	Gaussian, beta, gamma, binomial, Poisson, neg. binomial	–	–	Gaussian	beta, flexible beta, 0/1-inflated beta, ext. beta	Gaussian, binomial, multinomial, Poisson, neg. binomial, gamma	Gaussian, binomial, beta-binomial
HB spatial effects	ICAR, BYM, BYM2, Leroux, SAR, SLM	–	–	–	ICAR	ICAR, BYM2, Leroux	ICAR, BYM2
Benchmarking	ratio, difference, MSE-weighted, logit; one or two levels	–	raking, ratio, MSE-adjusted; one total	linear constraints	raking, ratio, double	–	national totals
Maps / export	yes / spreadsheet	– / –	yes / spreadsheet	– / –	yes / CSV	– / –	yes / –
Compiled code	C++ (OpenMP)	–	–	–	Stan	C++	–

of NumPyro (Hoffman and Gelman, 2014; Phan et al., 2019) through interchangeable engine objects. For beta, gamma and negative binomial data the dispersion of each area is fixed by the sampling variance of its direct estimate, and the default priors are documented together with the simulations that led to them.

3. Post-estimation functions for benchmarking at one or two levels, reliability classification, model comparison, maps and export, which accept the output of every model.

The remainder of the paper describes the methods and their implementation, reports benchmarks and a design-based simulation, and illustrates a complete workflow.

2 Methods

Throughout, areas are indexed by $d = 1, \ldots, D$, $\boldsymbol{x}_d$ is a p-vector of area-level covariates, and ψ_d is the design-based sampling variance of the direct estimate y_d of the area parameter θ_d.

2.1 Frequentist models

The Fay-Herriot model (Fay and Herriot, 1979) is

$$y_d = \boldsymbol{x}_d^\top \boldsymbol{\beta} + u_d + e_d, \quad u_d \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_u^2), \quad e_d \overset{\text{ind}}{\sim} \mathcal{N}(0, \psi_d), \tag{1}$$

with ψ_d known. Its empirical best linear unbiased predictor (EBLUP) is $\hat{\theta}_d = \hat{\gamma}_d y_d + (1 - \hat{\gamma}_d)\boldsymbol{x}_d^\top \hat{\boldsymbol{\beta}}$ with $\hat{\gamma}_d = \hat{\sigma}_u^2/(\hat{\sigma}_u^2 + \psi_d)$, where $\hat{\boldsymbol{\beta}}$ is the generalized least squares estimator at the REML or ML estimate $\hat{\sigma}_u^2$. The MSE is estimated by the second-order approximation $g_{1d} + g_{2d} + 2g_{3d}$ of Prasad and Rao (1990) for REML, with the bias correction of Datta and Lahiri (2000) for ML, or by parametric bootstrap.

The spatial Fay-Herriot model replaces the independent area effects by a simultaneous autoregressive (SAR) process, $\boldsymbol{u} = \rho \boldsymbol{W}\boldsymbol{u} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \sigma_u^2 \boldsymbol{I})$ and a row-standardized proximity matrix $\boldsymbol{W}$ (Petrucci and Salvati, 2006; Pratesi and Salvati, 2008). The marginal covariance is then $\boldsymbol{V} = \boldsymbol{G} + \boldsymbol{\Psi}$ with $\boldsymbol{G} = \sigma_u^2[(\boldsymbol{I} - \rho \boldsymbol{W})^\top (\boldsymbol{I} - \rho \boldsymbol{W})]^{-1}$ and $\boldsymbol{\Psi} = \operatorname{diag}(\psi_d)$, and the EBLUP is $\hat{\boldsymbol{\theta}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} + \hat{\boldsymbol{G}}\hat{\boldsymbol{V}}^{-1}(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})$. The process is defined on all areas of $\boldsymbol{W}$, so areas without a sample are predicted as well. (σ_u^2, ρ) are estimated by REML or ML Fisher scoring on $|\rho| \leq 0.999$, with a projected iteration when the unconstrained one ends at a negative variance. MSEs follow Molina and Marhuenda (2015) (analytical, parametric or non-parametric bootstrap).

For repeated surveys, $t = 1, \ldots, T$, the spatio-temporal model of Marhuenda et al. (2013) is

$$y_{dt} = \boldsymbol{x}_{dt}^{\top}\boldsymbol{\beta} + u_{1d} + u_{2dt} + e_{dt}, \tag{2}$$

with a SAR area effect u_{1d} and an AR(1) effect $u_{2dt} = \rho_2 u_{2d,t-1} + \varepsilon_{2dt}$ within each area. The panel need not be complete: the model is defined on the full $D \times T$ grid, the variance parameters are estimated by REML from the observed cells, and missing periods and areas that were never sampled are predicted from the observed ones (Saei and Chambers, 2005), with analytical and parametric bootstrap MSEs for every cell.

At the unit level, the nested error model of Battese et al. (1988), $y_{dj} = \boldsymbol{x}_{dj}^{\top}\boldsymbol{\beta} + u_d + e_{dj}$ for unit j of area d, gives the EBLUP of the area mean $\hat{\bar{Y}}_d = f_d \bar{y}_d + (1 - f_d)\{\bar{\boldsymbol{X}}_{d,r}^{\top}\hat{\boldsymbol{\beta}} + \hat{\gamma}_d(\bar{y}_d - \bar{\boldsymbol{x}}_d^{\top}\hat{\boldsymbol{\beta}})\}$, where $f_d = n_d / N_d$ is the sampling fraction and $\bar{\boldsymbol{X}}_{d,r}$ the covariate mean of the non-sampled units. `eblup_twofold()` fits the two-fold model with area and sub-area effects in the same way.

2.2 Hierarchical Bayes models

Latent Gaussian models

The HB functions formulate each model as a latent Gaussian model (Rue et al., 2009). For `hb_area()`,

$$y_d \mid \mu_d \sim \pi(y_d \mid \mu_d, \phi_d), \qquad g(\mu_d) = \boldsymbol{x}_d^{\top}\boldsymbol{\beta} + u_d, \tag{3}$$

where g is the identity, logit or log link of the family, ϕ_d a dispersion parameter (below), and $\boldsymbol{u}$ a zero-mean Gaussian Markov random field: independent effects or one of the spatial structures below, plus temporal and space-time terms for panel data. `hb_unit()` applies the same construction to unit records with an area effect, and `hb_twofold()` to sub-areas within areas. The INLA engine approximates the posterior marginals by nested Laplace approximations and numerical integration over the hyperparameters; **fastsae** uses the full Laplace approximation by default, which is more accurate than the simplified one for non-Gaussian likelihoods with little information per area. The JAX engine samples the same model with the No-U-Turn sampler.

Dispersion from the sampling variances

When the sampling variances ψ_d are known, **fastsae** fixes the dispersion of each area so that the model variance of y_d equals ψ_d:

- **Gaussian**: the observation variance is ψ_d, which gives the Fay-Herriot model (1).
- **Beta**: for proportions with a logit link, $\mathrm{Var}(y_d) = \mu_d(1-\mu_d)/(1+\phi_d)$, which gives the precision of Janicki (2020),
$$\phi_d = \max\left\{\frac{\mu_d(1-\mu_d)}{\psi_d} - 1,\, 1\right\}. \tag{4}$$
It depends on the unknown μ_d. Replacing μ_d by y_d makes the model variance $\psi_d\mu_d(1-\mu_d)/\{y_d(1-y_d)\}$, too small wherever y_d is far from μ_d; in a simulation with proportions near 0.08 the 95% intervals then covered 81–84% of the true values. **fastsae** therefore fits the model twice and evaluates (4) at the posterior means of the first fit, which gave 90–94%. **tipsae** instead makes ϕ_d a function of μ_d inside the model (De Nicolò and Gardini, 2024); this covered 93–95% in the same simulation, but makes the likelihood bimodal, with a mode of low precision near zero, and 11 of 240 fits did not converge ($\hat{R} > 1.05$). With known sample sizes instead of variances, $\phi_d = n_d - 1$.
- **Gamma**: for positive skewed indicators with a log link, $\mathrm{Var}(y_d) = \mu_d^2/s_d$ gives the shape $s_d = y_d^2/\psi_d$, the inverse squared coefficient of variation.
- **Negative binomial**: $\mathrm{Var}(y_d) = \mu_d + \mu_d^2/k_d$ gives the size $k_d = \hat{\mu}_d^2/(\psi_d - \hat{\mu}_d)$, again at the posterior means of a first fit.
- **Binomial and Poisson**: sample sizes are the numbers of trials, and population exposures enter as an offset.

Without sampling variances, one dispersion parameter is estimated. With one direct estimate per area it trades off against the area effect, so its prior determines the intervals (Table 2), and `hb_area()` warns.

Spatial and space-time effects

The area effect supports the intrinsic conditional autoregressive (ICAR) model of Besag et al. (1991), the BYM model with independent and ICAR components, the scaled BYM2 model $u_d = \sigma_u(\sqrt{1-\phi}\, v_d + \sqrt{\phi}\, u_d^*)$ of Riebler et al. (2016), where u^* is a scaled ICAR field, the Leroux model with precision $\tau[(1-\lambda)\boldsymbol{I} + \lambda(\mathrm{diag}(\boldsymbol{W}\boldsymbol{1}) - \boldsymbol{W})]$ (Leroux et al., 2000), and SAR models for the area effect, as in the spatial Fay-Herriot model, or as a spatial lag model. Temporal effects can be random walks of order one or two, AR(1) or independent, and space-time interactions can be area-specific trends, a spatial field with AR(1) dynamics, or one of the four types of Knorr-Held (2000).

Default priors

With few areas, or one observation per area, the data say little about the variance and dispersion parameters, and their priors decide how strongly the estimates are shrunk and how wide the intervals are. Table 2 lists the defaults, which both engines share.

Table 2: Default priors of hb_area(), hb_unit() and hb_twofold(). All intrinsic Gaussian Markov random fields are scaled, so that σ is a typical marginal standard deviation for every structure. Every prior can be changed through inla_engine().

Parameter	Prior
Coefficients $\boldsymbol{\beta}$	intercept flat; others $\mathcal{N}(0, 1000)$
Standard deviations of random effects (area, sub-area, time, space-time; σ_e of hb_unit())	half-normal(s), $s = \mathrm{sd}(\boldsymbol{y})$ for Gaussian models and $s = 1$ on the link scale otherwise
BYM2 mixing parameter ϕ	PC prior with $P(\phi < 0.5) = 0.5$
Leroux λ; SAR and SLM ρ	uniform on the admissible range
AR(1) correlation ρ	$\mathcal{N}(0, 1/0.15)$ on $\log\{(1+\rho)/(1-\rho)\}$
Gamma shape, negative binomial size (no ψ_d)	half-normal(10)
Beta precision (no ψ_d, no n_d)	exponential with rate 0.1
Gaussian residual standard deviation (no ψ_d)	half-normal(s)

The half-normal prior is a weakly informative choice for hierarchical standard deviations (Gelman, 2006). Its scale is tied to the response, because a penalized complexity (PC) prior $P(\sigma > U) = \alpha$ (Simpson et al., 2017) with the common $U = 1$ depends on the unit of y: in a Fay-Herriot simulation with 30 areas the 95% intervals under PC$(1, 0.01)$ covered 94%, 85% and 38% of the true means for the same data in three units. The scaled half-normal prior covered as well as the EBLUP intervals (93–97% for 15 to 60 areas, against 93–96%), whereas PC priors with $U = \mathrm{sd}(\boldsymbol{y})$ covered 85–94%, because they put much mass near $\sigma = 0$ (Sørbye and Rue, 2014). On the logit and log scales, half-normal(1) covered at least as well as the half-normal(2.5) default of **tipsae** and estimated the area standard deviation with less bias. For the dispersion parameters estimated without ψ_d, the INLA defaults put the gamma shape near 80 and the negative binomial size near 110–180 whatever the truth, and intervals for overdispersed data covered 52–92% of the true means; the half-normal(10) prior covered 90–99%, at the cost of too wide intervals when the sampling noise is small, which is why ψ_d should be supplied whenever it is known. These simulations and their code are documented in `vignette("hb-priors")`.

2.3 Benchmarking

Small area estimates $\hat{\theta}_d$ are often required to agree with direct estimates of larger domains (Rao and Molina, 2015; Stefan and Hidiroglou, 2021). Let areas d belong to groups g (provinces, say) with weights w_d (population shares) normalized within groups, $\tilde{w}_d = w_d / \sum_{d' \in g} w_{d'}$, and let T_g be the group targets. `benchmark_sae()` adjusts the estimates so that $\sum_{d \in g} \tilde{w}_d \hat{\theta}_d^{\mathrm{B}} = T_g$ by

- ratio adjustment, $\hat{\theta}_d^{\mathrm{B}} = \hat{\theta}_d T_g / \sum_{d' \in g} \tilde{w}_{d'} \hat{\theta}_{d'}$ (You and Rao, 2002);
- difference adjustment, $\hat{\theta}_d^{\mathrm{B}} = \hat{\theta}_d + T_g - \sum_{d' \in g} \tilde{w}_{d'} \hat{\theta}_{d'}$;
- MSE-weighted adjustment, which minimizes $\sum_d (\hat{\theta}_d^{\mathrm{B}} - \hat{\theta}_d)^2 / \mathrm{MSE}(\hat{\theta}_d)$ under the constraint and

so moves imprecise estimates most (Datta et al., 2011; Bell et al., 2013):

$$\hat{\theta}_d^{\mathrm{B}} = \hat{\theta}_d + \frac{\tilde{w}_d \,\mathrm{MSE}(\hat{\theta}_d)}{\sum_{d' \in g} \tilde{w}_{d'}^2 \,\mathrm{MSE}(\hat{\theta}_{d'})} \Big(T_g - \sum_{d' \in g} \tilde{w}_{d'} \hat{\theta}_{d'} \Big); \tag{5}$$

- a shift α_g on the logit scale, $\sum_{d \in g} \tilde{w}_d \,\mathrm{logit}^{-1}\{\mathrm{logit}(\hat{\theta}_d) + \alpha_g\} = T_g$, solved by one-dimensional root finding, which keeps proportions inside $(0, 1)$.

With a national target T as well, the group targets are first adjusted so that their weighted mean equals T, and the areas are then benchmarked to the adjusted group targets. hb_area() and eblup_fh() can also benchmark within the fit (argument `self_benchmark`); for hb_area(), the posterior draws are then benchmarked, so the intervals refer to the benchmarked estimates.

2.4 Reliability classification

Statistical offices publish small area estimates by tiers of their relative standard error, $\mathrm{RSE}_d = \sqrt{\mathrm{MSE}(\hat{\theta}_d)}/|\hat{\theta}_d|$. `add_reliability_flags()` labels an estimate reliable for $\mathrm{RSE}_d \leq c_1$, to be used with caution for $c_1 < \mathrm{RSE}_d \leq c_2$, and unreliable above c_2. The defaults $c_1 = 25\%$ and $c_2 = 50\%$ are those of BPS-Statistics Indonesia; other cut-offs can be supplied.

3 Implementation

Table 3 lists the main functions. The frequentist functions call C++ code (**RcppArmadillo**, **RcppEigen**) that never forms a dense inverse of the marginal covariance matrix. The spatial model works with sparse Cholesky factors of $(\boldsymbol{I} - \rho \boldsymbol{W})^\top (\boldsymbol{I} - \rho \boldsymbol{W})$ and of $(\boldsymbol{I} - \rho \boldsymbol{W})^\top (\boldsymbol{I} - \rho \boldsymbol{W}) + \sigma_u^2 \boldsymbol{\Psi}^{-1}$ through the Woodbury identity; the spatio-temporal and two-fold models compute REML scores, Fisher information and the analytical MSE from $T \times T$, $D \times D$ and low-rank blocks; and the unit-level model works with area-level sufficient statistics, profiling $\boldsymbol{\beta}$ and σ_e^2 out of the (restricted) likelihood. Bootstrap replicates run in parallel with OpenMP; their random numbers are drawn beforehand from R's generator, so the results depend on the seed but not on the number of threads.

The HB functions translate formulas, adjacency information and sampling variances into a model and pass it to an engine object. `inla_engine()` (the default) holds the INLA strategy, the priors and further arguments of `INLA::inla()`; `jax_engine()` holds the device (CPU or GPU), the number of chains and draws and the sampler settings, and runs NumPyro through **reticulate**. Both engines return the same object, so `print()`, `summary()`, `plot()`, `coef()`, `fitted()` and the post-estimation functions do not depend on how a model was estimated. INLA is not on CRAN and Python is optional, so both are suggested dependencies, as are **sf** for maps and a spreadsheet writer for export; the frequentist and post-estimation functions need neither INLA nor Python.

Table 3: Main functions of fastsae.

Function	Purpose
eblup_fh(), eblup_sfh(), eblup_stfh()	Fay-Herriot, spatial and spatio-temporal EBLUP with analytical or bootstrap MSE
eblup_bhf(), eblup_twofold()	Unit-level and two-fold EBLUP
hb_area()	Area-level HB models: six likelihoods, spatial, temporal and space-time effects
hb_unit(), hb_twofold()	Unit-level and two-fold HB models
inla_engine(), jax_engine()	Estimation engines for the HB functions
benchmark_sae()	Ratio, difference, MSE-weighted and logit benchmarking at one or two levels
add_reliability_flags()	Reliability tiers by relative standard error
compare_sae(), step_sae()	Agreement and ranking of two models; covariate selection
diagnose(), autoplot()	Model diagnostics and plots
map_sae(), export_sae()	Choropleth maps; spreadsheet export

4 Benchmarks and simulations

All timings were measured on an Apple M4 (10 cores, 16 GB) under R 4.6.1. The scripts are in the `benchmarks/` directory of the package repository and store their results in `inst/extdata/`, from which this article computes its tables and figures.

4.1 Frequentist models

Table 4: Median time and memory allocated by R for point estimates and analytical MSEs. The spatio-temporal model has $T = 5$ periods. The fastsae rows were measured with fastsae 1.0.0; the sae and emdi rows come from an earlier run of the same design on other random draws.

	Time (s)			Memory allocated (MB)		
D	fastsae	sae	emdi	fastsae	sae	emdi
Fay-Herriot						
30	0.00034	0.0016	0.01	0.42	0.15	4.18
100	0.00043	0.0034	0.05	0.04	0.92	37.34
500	0.00073	0.20	5.86	0.21	18.24	880
1,000	0.0012	1.50	51.02	0.41	71.98	3,774
Spatial Fay-Herriot						
30	0.04	0.0041	0.01	0.04	1.37	4.18
100	0.12	0.07	0.05	0.16	17.98	37.34
500	1.76	7.12	3.14	3.09	776	950
1,000	8.70	67.15	51.06	11.90	1,920	3,773
Spatio-temporal Fay-Herriot						
30	0.14	0.12	–	2.72	43.80	–
100	0.17	17.94	–	0.56	2,551	–
500	7.34	253	–	5.13	8,124	–
1,000	50.53	1,786	–	15.96	32,478	–

We timed the three area-level models with point estimates and analytical MSEs on simulated data with five covariates and D up to 1,000 areas (Table 4). Times are medians from the **bench** package, and memory is the total memory allocated by R during one call (`mem_alloc`), not the peak memory of the process, and does not include memory allocated by compiled code. The rows for **sae** and **emdi** come from an earlier run of the same design whose data generator is no longer available, so they were obtained on other random draws than the **fastsae** rows; we report them for orientation. At $D = 1{,}000$, **fastsae** fitted the Fay-Herriot model in 0.0012 s against 1.50 s for **sae**, the spatial model in 8.70 s against 67.15 s, and the spatio-temporal model in 50.53 s against 30 minutes, for which **sae** allocated 31.7 GB. For small D the fixed costs dominate and **sae** is as fast or faster for the spatial models.

4.2 Beta models: fastsae and tipsae

Table 5: Median time in seconds of beta models with sampling variances, fitted by fastsae (INLA, two-stage precision) and tipsae (Stan, 4 chains of 2,000 iterations). The spatio-temporal model has $T = 5$ periods and was run up to $D = 50$.

	Independent		Spatial (ICAR)		Spatio-temporal	
D	fastsae	tipsae	fastsae	tipsae	fastsae	tipsae
30	2.85	2.31	2.66	48.71	2.93	574
50	2.56	4.35	2.51	55.15	3.80	716
100	2.73	10.94	2.67	89.36	–	–
250	3.27	33.58	2.75	145	–	–
500	3.31	82.75	3.03	232	–	–
1,000	3.73	180	4.28	362	–	–

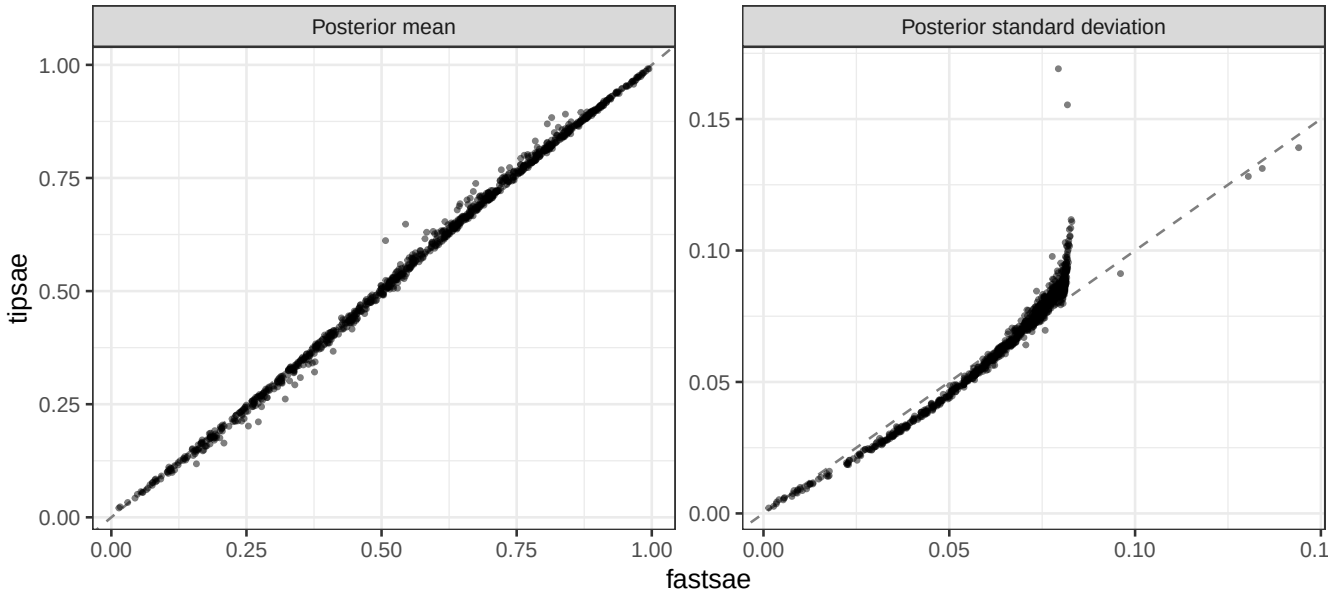


Figure 1: Posterior means (left) and posterior standard deviations (right) of a beta model with $D = 1{,}000$ areas fitted by fastsae and tipsae. The dashed lines are the identity.

For beta models we compared `hb_area(family = "beta")` with **tipsae** (De Nicolò and Gardini, 2024) on data simulated by `sim_area_data()`, with sampling variances $\psi_d = y_d(1 - y_d)/26$ computed from the direct estimates as a survey would, and **tipsae** with `type_disp = "var"` and its default sampler settings (Table 5). The two packages do not fit the same model: their priors differ (see Default priors), and the precision is fixed at a first-stage estimate in **fastsae** but a function of the model mean in **tipsae**. The comparison is therefore between what a user obtains from each package with its defaults. At $D = 1{,}000$ the independent model took 3.73 s in **fastsae** and 180 s in **tipsae**, and the spatial model 4.28 s and 362 s. At these sizes most of an INLA fit is fixed cost, such as starting the INLA program, and the two-stage precision fits the model twice, so the runtime of **fastsae** grows slowly with D.

To compare the estimates, we fitted both packages to the $D = 1{,}000$ data set of Table 5 and compared them with the true area means (Figure 1). The posterior means of the two packages correlated at $r = 0.9988$ with a mean absolute difference of 0.0096, and the posterior standard deviations of **fastsae** were 0.99 times those of **tipsae** in the median area (90% of the areas between 0.88 and 1.17). Against the true means, the RMSE was 0.0678 for **fastsae** and 0.0684 for **tipsae**, and the 95% intervals covered 93.3% and 91.1% of the true means, with mean widths 0.252 and 0.259.

4.3 Estimation engines

Table 6: Time in seconds of a second (warm) fit of hb_area() with an ICAR area effect, by estimation engine. JAX: NumPyro NUTS, 2 chains of 1,000 draws after 500 warm-up iterations; tipsae: Stan defaults (4 chains of 2,000 iterations), beta model only. Times are for these run lengths, not per effective sample.

D	INLA	JAX (CPU)	JAX (Apple GPU)	tipsae
Gaussian				
100	0.73	4.51	136	–
500	3.12	32.48	245	–
1,000	0.80	92.72	377	–
2,500	1.41	841	725	–
Beta				
100	2.60	24.00	221	66.73
500	3.11	146	281	152
1,000	3.72	374	376	235
2,500	6.32	1,269	819	620

The same `hb_area()` model can be estimated by INLA or by the No-U-Turn sampler of NumPyro

on JAX, on the CPU or a GPU. Table 6 compares the engines on Gaussian and beta models with an ICAR area effect on a regular grid with queen adjacency; for the beta model, **tipsae** was run with its ICAR effect. Each engine ran in its own R process and each fit was run twice on the same data. At $D = 2,500$ INLA fitted the spatial beta model in 6.32 s, against 620 s for **tipsae** and 1,269 s for JAX on the CPU. The JAX engine is a way to check INLA by MCMC rather than a faster alternative to it. Its time grows faster with D than that of INLA, because it represents the ICAR effect through the eigenvectors of the graph Laplacian, which costs $O(D^2)$ per gradient, and for the beta model it runs the sampler twice, once for each stage of the precision. On the CPU it took 0.4 times the time of **tipsae** at $D = 100$ and 2.0 times at $D = 2,500$. The Apple GPU has a fixed cost of about 2 minutes per sampler run, as each step of the tree-building loop of NUTS synchronizes GPU and CPU; it pays off only for the largest models. Running NumPyro on Apple GPUs required replacing two operations that **jax-metal** cannot compile (`popcnt` and `lax.cond` with constant outputs). The posterior means of JAX correlated with those of INLA at $r \geq 0.9999$ with median ratios of posterior standard deviations between 1.00 and 1.02. The largest split $\hat{R}$ of the JAX fits was 1.016, slightly above the threshold of 1.01 recommended by Vehtari et al. (2021), so longer chains than these two of 1,000 draws are advisable for inference. Not all **tipsae** fits converged: the largest $\hat{R}$ was 3.71, at $D = 100$, in line with the bimodal likelihood described under Dispersion from the sampling variances, so its times refer to the run length and not to a converged posterior.

4.4 Design-based simulation

To assess the estimators under repeated sampling from a finite population, we generated a population of about $N = 250,000$ individuals in $D = 500$ areas on a 20×25 grid, with a binary outcome that is logistic in two covariates with ICAR and independent area effects, and drew $R = 25$ samples of 20 to 50 units per area by simple random sampling. Each sample was fitted by a beta model with an ICAR area effect and the sampling variances of the direct proportions, by `hb_area()` with its defaults and by **tipsae** (`type_disp = "var"`, one chain of 500 iterations after 150 warm-up iterations). The script is `benchmarks/sim_mc_finite_pop_d500.R`.

Table 7: Design-based simulation ($D = 500$, $R = 25$): root mean squared error, relative RMSE and mean absolute relative bias, averaged over areas, coverage of the 95% intervals (standard error across areas in parentheses), mean interval width, and time per fit with five replicates running in parallel.

Estimator	RMSE	RRMSE (%)	MARB (%)	Coverage (%)	Width	Time (s)
Direct	0.0719	27.1	21.7	–	–	–
fastsae	0.0489	18.8	15.8	93.4 (0.5)	0.194	8.5
tipsae	0.0487	18.7	15.7	93.7 (0.4)	0.202	40.0

Both HB fits reduced the RMSE of the direct estimator from 0.0719 to about 0.049, an efficiency gain (ratio of mean squared errors) of 1.99 (Table 7). The two packages were equivalent in accuracy and in the coverage of their 95% intervals, which was slightly below the nominal level for both; the intervals of **fastsae** were slightly narrower. An earlier version of this simulation, with a PC prior on the area standard deviation and the precision evaluated at the direct estimates, covered 89.0%, against 93.4% with the default priors and the two-stage precision.

5 Example: poverty in Emilia-Romagna

We illustrate a complete workflow with the head-count poverty ratios of the 38 health districts of the Italian region Emilia-Romagna in 2016, distributed with **tipsae** (De Nicolò and Gardini, 2024). The data contain the direct estimates `hcr`, their sampling variances `vars`, the sample sizes `n` (10 to 285 households), the populations `pop` and the province `prov` of each district, and a shapefile of the districts. The only covariate, `x`, is artificial, so the example illustrates the software rather than a substantive analysis.

```
library("fastsae")
data("emilia_cs", "emilia_shp", package = "tipsae")
shp <- sf::st_as_sf(emilia_shp)
shp <- shp[match(emilia_cs$id, shp$NAME_DISTRICT), ]
W <- spdep::nb2mat(spdep::poly2nb(shp), style = "B")
dimnames(W) <- list(emilia_cs$id, emilia_cs$id)
```

We fit a Fay-Herriot model and a beta model with a BYM2 area effect, both with the sampling variances of the direct estimates:

```
fit_fh <- eblup_fh(hcr ~ x, vardir = "vars", domain = "id",
  data = emilia_cs, print_result = FALSE)
fit_beta <- hb_area(hcr ~ x, data = emilia_cs, domain = "id",
  vardir = "vars", family = "beta", spatial = "bym2", W = W,
  print_result = FALSE)
```

`compare_sae()` summarizes how far the two sets of estimates agree and which of them has the smaller estimated uncertainty; below are its agreement and efficiency measures:

```
cmp <- compare_sae(fit_fh, fit_beta, names = c("FH", "Beta-BYM2"))
subset(cmp$metrics, Group %in% c("Agreement", "Efficiency"), c(Metric, Value))
```

```
#>                                                    Metric             Value
#> 2                          Mean Difference (Beta-BYM2 - FH)            0.0013
#> 3                                   95% Limits of Agreement [-0.0110, 0.0137]
#> 4                            Mean Absolute Difference (MAE)            0.0042
#> 5                     Root Mean Squared Difference (RMSD)            0.0064
#> 6                     Lin's Concordance Correlation (CCC)            0.9604
#> 13                         Mean MSE Ratio (FH / Beta-BYM2)            1.1486
#> 14                       Median MSE Ratio (FH / Beta-BYM2)            1.1611
#> 15 % of domains where Beta-BYM2 reports lower uncertainty             63.2%
#> 16                                        Mean RSE % (FH)            16.58%
#> 17                                  Mean RSE % (Beta-BYM2)            15.35%
```

The regional poverty ratio, estimated directly from the regional sample, is precise enough to publish, so we benchmark the district estimates to it, and to the provincial direct estimates, with the logit adjustment that keeps them in $(0, 1)$. For the illustration we use population-weighted means of the district direct estimates as targets:

```
prov_target <- tapply(emilia_cs$hcr * emilia_cs$pop, emilia_cs$prov, sum) /
  tapply(emilia_cs$pop, emilia_cs$prov, sum)
reg_target <- sum(emilia_cs$hcr * emilia_cs$pop) / sum(emilia_cs$pop)
bm <- benchmark_sae(fit_beta, target = prov_target, group = emilia_cs$prov,
  weight = emilia_cs$pop, national_target = reg_target, method = "logit")
head(bm[, c("domain", "group", "original", "benchmarked")], 4)
```

```
#>                domain group   original benchmarked
#> 1 CASALECCHIO DI RENO    BO 0.05724873  0.05642077
#> 2   CITTA' DI BOLOGNA    BO 0.06635293  0.06540243
#> 3               IMOLA    BO 0.07000447  0.06900553
#> 4         PIANURA EST    BO 0.06648964  0.06553732
```

Finally, the estimates are classified by their RSE, mapped (Figure 2), and exported to a spreadsheet with `export_sae()`:

```
as.data.frame(table(tier = add_reliability_flags(fit_beta)$reliability))
```

```
#>                                tier Freq
#> 1              Reliable (RSE <= 25%)   38
#> 2 Use with caution (25% < RSE <= 50%)    0
#> 3            Unreliable (RSE > 50%)    0
```

```
p_est <- map_sae(fit_beta, sf_geom = shp, key = "NAME_DISTRICT",
  type = "estimate", title = "Head-count ratio", subtitle = "")
p_rse <- map_sae(fit_beta, sf_geom = shp, key = "NAME_DISTRICT",
  type = "rse", title = "RSE (%)", subtitle = "")
patchwork::wrap_plots(p_est, p_rse)
```

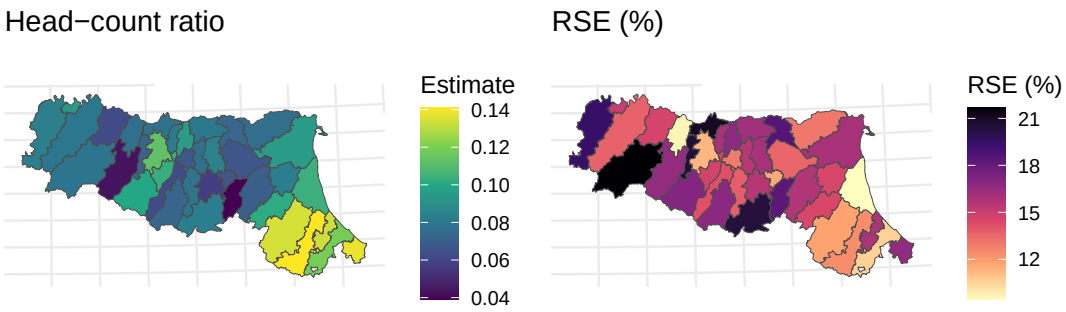


Figure 2: Head-count poverty ratios of the health districts of Emilia-Romagna in 2016 estimated by the beta model with a BYM2 area effect (left) and their relative standard errors (right).

6 Summary and discussion

fastsae brings the frequentist and hierarchical Bayes small area models used in official statistics, and the steps that follow a fit, together in one interface. The compiled frequentist implementations reduce the time of the spatio-temporal Fay-Herriot model with analytical MSE at 1,000 areas from half an hour to about a minute, and its memory allocated by R from gigabytes to megabytes. The HB functions fit beta and other non-Gaussian models in seconds with INLA, and the same models can be checked against full MCMC with the JAX engine without rewriting them.

Some limitations should be kept in mind. INLA has a fixed cost per fit, so for small Gaussian models numerical integration as in **hbsae** is faster: for the Fay-Herriot model with 250 areas, `hb_area()` took 0.70 s and `hbsae::fSAE.Area()` 0.05 s (script `benchmarks/bench_hb_normal.R`). The beta and negative binomial dispersions are fixed at first-stage posterior means, and their uncertainty is not propagated. INLA is an approximation whose accuracy can deteriorate for non-Gaussian likelihoods with very little information per area; the JAX engine allows such cases to be checked. Our benchmarks were run on a single machine, and the runtimes of **sae** and **emdi** were not measured on the same draws as those of **fastsae**.

Future work will add unit-level models for non-Gaussian outcomes, zero-inflated beta models for indicators with many zeros, and benchmarking that propagates into the MSE of frequentist estimates.

Computational details

The results were obtained with R 4.6.1, **fastsae** 1.0.0, **R-INLA** 26.8.7, **tipsae** 1.0.3, **sae** 1.3, **emdi** 2.2 and **hbsae** 1.2; the JAX engine used JAX 0.5.0 and NumPyro 0.19.0. R and the packages are available from CRAN at `https://CRAN.R-project.org/`, **R-INLA** from `https://www.r-inla.org/`, and the source of **fastsae**, including the benchmark scripts, from `https://github.com/ridsonap/fastsae`.

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